

Nonlinear physics of moiré superlattices

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Nonlinear physics is one of the most important research fields in modern physics and materials science. It offers an unprecedented paradigm for exploring many fascinating physical phenomena and realizing diverse cutting-edge applications inconceivable in the framework of linear processes. Here we review the recent theoretical and experimental progress concerning the nonlinear physics of synthetic quantum moiré superlattices. We focus on the emerging nonlinear electronic, optical and optoelectronic properties of moiré superlattices, including but not limited to the nonlinear anomalous Hall effect, dynamically twistable harmonic generation, nonlinear optical chirality, ultralow-power-threshold optical solitons and spontaneous photogalvanic effect. We also present our perspectives on the future opportunities and challenges in this rapidly progressing field, and highlight the implications for advances in both fundamental physics and technological innovations.

Nonlinear physics encompasses the study of material responses that depend on the strength of external stimuli (such as applied electric or optical fields) in a nonlinear manner. It constitutes a pivotal research domain within condensed-matter physics, materials science and modern optics (Box 1)^{1–5}. Typically, nonlinear responses of a material are more susceptible to its physical properties, including crystal symmetry, electronic structure and non-equilibrium carrier distributions, than its linear responses. This manifests nonlinear effects as an unparalleled paradigm to probe the materials' underlying physics and quantum phase transitions, which are hard to detect or even unattainable with linear processes, such as hidden-order phase, spin–orbit interaction, ultrafast electron dynamics, ferroelectricity, magnetism, superconductivity, band geometry and topology^{3–5}. For instance, second-harmonic generation (SHG) within the dipolar approximation can occur only in non-centrosymmetric materials, and therefore offers an ideal technique to detect spontaneous inversion symmetry breaking and order parameters of ferroelectric metals that are inaccessible

by the conventional linear means⁵. In contrast to linear Hall responses constrained in systems with broken time-reversal symmetry (such as materials under an external magnetic field or magnetic materials), nonlinear Hall effects (NHEs) can go beyond such restrictions and present new avenues to probe the Berry curvature dipole (BCD) and topological quantum phase transitions for non-magnetic materials under zero magnetic field³. Additionally, nonlinearities lay a solid foundation for a wide range of applications, such as ultrafast lasers, all-optical information processing, attosecond science, imaging, energy harvesting, quantum information technologies and multipetahertz and lightwave electronics^{1,2}.

Triggered by the ground-breaking discoveries of correlated insulating states and superconductivity in magic-angle twisted bilayer graphene (TBG)^{6,7}, moiré superlattices formed by superimposing two or more periodic lattices with a slight lattice mismatch and/or a rotation angle have attracted widespread interest worldwide and opened a new research frontier of moiré physics^{8–11}. In the span

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BOX 1

Fundamentals of nonlinear physics

The responses of a material to an applied optical field $\mathbf{E}$ can be described by a power-series expansion of the dielectric polarization $\mathbf{P}$ in terms of the field strength:

$$\mathbf{P} = \epsilon_0 [\chi^{(1)}\mathbf{E} + \chi^{(2)}\mathbf{E}^2 + \chi^{(3)}\mathbf{E}^3 + \chi^{(4)}\mathbf{E}^4 + \dots] \quad (1)$$

where ϵ_0 is the vacuum permittivity and $\chi^{(n)}$ is the n th-order susceptibility tensor with rank $(n+1)$, hinging on the point group of the material. The first term in equation (1) describes the conventional linear effects, such as absorption, refraction and scattering. The higher-order ($n \geq 2$) terms in equation (1) correspond to nonlinear responses, typically producing new electromagnetic radiation at the sums and differences of the incident frequencies.

Nonlinear responses are usually much smaller than linear ones and therefore require an intense field. Indeed, the discovery of SHG in 1961, which marked the birth of nonlinear optics, benefited directly from the invention of the laser almost one year earlier. Apart from the celebrated SHG process, the second-order nonlinear response also contains sum- and difference-frequency generation. Specifically, difference-frequency generation corresponds to a non-oscillating d.c. response of zero frequency when only one input optical field is employed. In other words, a non-reciprocal static electric field, rather than electromagnetic radiation, is created. Such a process of zero frequency is known as optical rectification. One striking manifestation is the bulk photovoltaic effect (BPGE, also referred to as intrinsic/spontaneous photovoltaic effect). In light of the polarization state of the photons that induce the current, the BPGE can be categorized into CPGE and LPGE, which occur under circularly and linearly polarized irradiation, respectively. In

third-order nonlinear optical processes, third-harmonic generation (THG) and four-wave mixing are typical examples. Additionally, third-order response also involves electromagnetic radiation at the original frequency, such as the optical Kerr effect arising from the modification of permittivity by an optical field: $\epsilon = \epsilon_0 [\chi^{(1)} + 3\chi^{(3)}|\mathbf{E}|^2]$. High-order nonlinear responses typically described by susceptibility of $n \geq 5$ enable unique capabilities to underpin exotic physics in the non-perturbative or extreme nonlinear regime. For example, HHG has enabled the birth of attosecond science and initiated a rich variety of emergent technological applications, such as the generation of extreme ultraviolet/soft X-ray coherent radiation, on-chip ultrafast nanophotonics and electronic structure probing.

Besides the widely studied nonlinear optical physics, recent advances have demonstrated a radically new type of nonlinear response—nonlinear electric transport, characterized by a harmonic voltage signal nonlinearly dependent on the driving electric field. Nonlinear electric transport and nonlinear optics share many similar aspects. They follow the same symmetry constraints: for example, both even-order nonlinear optics and electric transport require broken inversion symmetry and can occur only in non-centrosymmetric materials. On the other hand, nonlinear electric transport differs from nonlinear optical effects in many aspects. For example, nonlinear optics are typically driven under high frequencies of $\sim 10^{14}$ Hz, and thus have a substantial contribution from interband transitions. By contrast, the emergent nonlinear electric transport is in the extremely low-driving-frequency regime (approximately 100 Hz) and therefore provides unprecedented possibilities to detect the spectral, symmetry, topological and geometrical properties of electron wavefunctions near the Fermi surface.

of just a few years, a rich variety of fascinating electrical, optical, thermal, magnetic, acoustic, mechanical and topological properties have emerged in moiré superlattices with an exceptional speed (Fig. 1), including but not limited to the fractional quantum anomalous Hall effect^{12–14}, Wigner crystal states^{15,16}, moiré excitons^{17–19}, moiré lasers²⁰, superb deformability²¹, condensed-matter quantum simulators^{9,22}, localization–delocalization transitions of light^{23,24} and intelligent infrared sensing²⁵. This delightful series of moiré discoveries has revolutionized our understanding of materials and reshaped the landscape of fundamental scientific research and technological innovation.

In the realm of nonlinear physics, moiré superlattices have emerged as a highly controllable quantum platform for exploring new phenomena and device architectures (Fig. 1). In this Review, we begin with an introduction to the fundamentals of nonlinear physics and synthetic quantum moiré superlattices. We then present an overview of the recent experimental and theoretical progress on emerging nonlinear electronic, optical, and optoelectronic properties of moiré superlattices, such as nonlinear electric transport, dynamically twistable SHG, high-harmonic generation (HHG), extremely-low-power-threshold optical solitons, nonlinear optical chirality, nonlinearity of quasiparticles and linear/circular photovoltaic effects (LPGE/CPGE). Finally, we conclude with perspectives on future research directions, conceptual new nonlinear device applications and challenges in this rapidly expanding field. We refer the reader to other recent reviews for more detailed discussions on moiré electric transport, photonics and optoelectronics with an emphasis on physics in the conventional linear regime^{8–11,26–29}.

Moiré superlattices

In moiré superlattices, formation of superlattice potentials with new length scales many times larger than the periodicity of parent lattices offers extraordinary opportunities to engineer the band structures, symmetry breaking, quantum geometry and topology of Bloch wavefunctions. This facilitates a diverse range of exotic electronic, thermal, mechanical, optical and optoelectronic properties that radically differ from (or at times are entirely absent from) their constituent materials. In this section, we briefly discuss the fundamental physical properties of moiré superlattices, which are typically categorized as moiré materials and structures.

Moiré materials

One spectacular example of moiré materials is magic-angle TBG (Fig. 2a). In 2011, flat bands with substantially enhanced electronic correlations, vanishing Fermi velocity and peaked density of states were theoretically predicted in TBG at a series of specific twist angles, which are now recognized as the magic angles (Fig. 2b)^{29,30}. Galvanized by this landmark prediction, Cao et al. successfully fabricated magic-angle TBG devices and reported the observation of correlated insulating states and unconventional superconductivity in 2018 (Fig. 2c)^{6,7}. These two ground-breaking discoveries pertaining to magic-angle TBG have attracted tremendous attention and sparked a ‘gold rush’ of exploring moiré physics. Over the past few years, a rich quantum phase diagram of strongly correlated behaviours originating from the interplay of Coulomb interactions, broken symmetry and topology in the flat minibands (such as quantum anomalous Hall states, unconventional ferroelectricity, charge-ordered states, quantum criticality, orbital

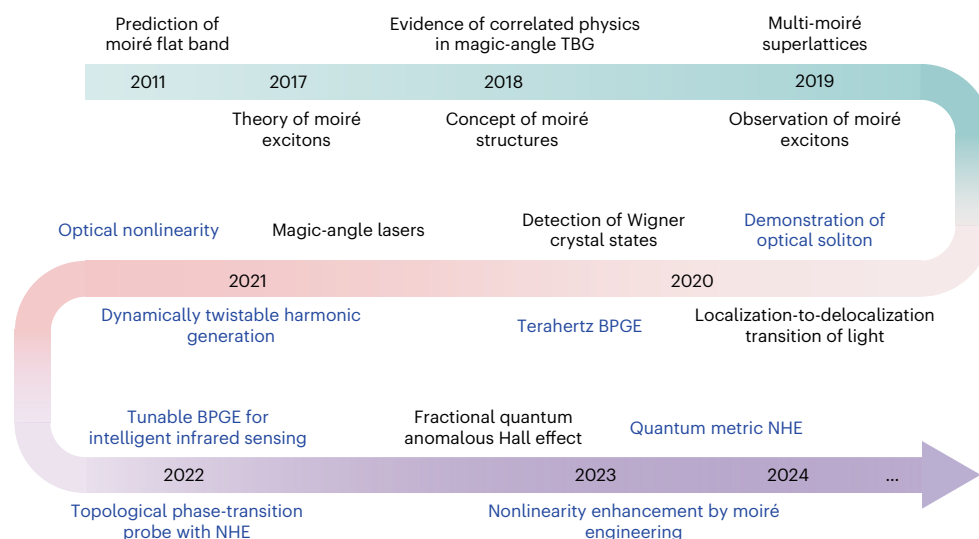


Fig. 1 | Timeline of selected moiré physics with an emphasis on nonlinear effects. Since the discovery of superconductivity and correlated insulating states in magic-angle TBG in 2018, a rich variety of fascinating electrical, optical, thermal, magnetic, acoustic and topological phenomena have emerged in

moiré materials and structures, with discoveries announced almost every week. The key developments in the nonlinear physics of moiré superlattices are highlighted in blue.

magnetism, fractional quantum spin Hall effect and cascades of phase transitions) has been witnessed in magic-angle TBG and a wide variety of other moiré materials, including, for example, twisted double-bilayer graphene (TDBG), angle-aligned ABC trilayer graphene/hexagonal boron nitride (h-BN) moiré superlattices and transition-metal dichalcogenide (TMD) homo- and hetero-bilayers^{12–14,28,29,31–36}. We envision that, despite remarkably rapid progress, only the tip of the iceberg has been uncovered thus far. More new moiré materials (such as twisted multilayer systems beyond single moiré, mixed- and hetero-dimensional moiré superlattices) will be created to possibly initiate more surprises, for example, fractional Chern insulators, non-Abelian topological order, quantum chiral spin liquids, Majorana fermions and room-temperature superconductivity.

Moiré structures

Inspired by the remarkable achievements of moiré materials, substantial endeavours have been devoted to designing moiré photonic and phononic structures. These synthetic moiré structures provide new paradigms to control electromagnetic, acoustic and elastic waves, and to simulate emergent quantum phenomena at the mesoscopic/macroscopic scale and in classical settings^{20,23,24,37–42}. For instance, moiré photonic structures formed by superimposing two honeycomb photonic crystals with a twist angle constitute a simple analogue of TBG, enabling the emergence of non-trivial topological transitions and dispersionless photonic bands ranging from the microwave to terahertz and visible frequencies (Fig. 2d,e)^{20,38,41,42}. Intriguingly, the photonic flat bands with zero light group velocities afford a new field confinement mechanism for high-quality nanocavities and laser physics. Indeed, a magic-angle nanolaser with remarkable figures of merit in terms of threshold, mode volume, lasing linewidth and operating temperature has been constructed based on moiré photonic structures by patterning two twisted sets of triangular lattices of nanoholes in the same semiconductor membrane (Fig. 2f)²⁰. Furthermore, reconfigurable coherent nanolaser arrays, a long-sought objective of laser research, have been developed based on moiré photonic flat bands, suggesting the prospect of applications in communication, computing, imaging and quantum nonlinear optics⁴³.

This successful analogue of moiré photonics to electronics can be extended to other physical systems, such as acoustics and mechanics

(Fig. 2g)^{37,40}. For example, moiré phononic structures emerging as a result of the interlayer coupling between two constituent phononic crystals facilitate the possibility of flat bands and non-Anderson-type localization of acoustic and elastic waves (Fig. 2h), affording new and exciting perspectives for understanding exotic quantum phenomena³⁷. Beyond mimicking moiré materials, moiré structures also show unique features. For example, in moiré structures the interlayer separation constitutes a novel degree of freedom to tailor the interlayer coupling strength^{38,40}. By reducing the interlayer separation, a strong interlayer coupling regime with flat bands appearing at magic angles larger than $\sim 6^\circ$ has been uncovered in both photonic and phononic moiré structures (Fig. 2i)^{38,40}. Notably, achieving such an extremely strong interlayer coupling in TBG requires pressures over ~ 130 GPa (ref. 44), well beyond the conventional experimental capability. We conceive that this exotic degree of freedom of interlayer separation in designing moiré photonic and phononic structures will introduce new promising paradigms for exploring extreme (for example, high-pressure) physics and emerging technological applications unattainable with moiré materials (for example, nanolasers, nonlinear cavity quantum electrodynamics, micro-light-emitting diodes, sensors and modulators).

State-of-the-art nonlinear physics of moiré superlattices

The artificial quantum moiré materials and photonic/phononic structures introduce a new length and energy scale, which provides a promising means of engineering the electronic/electromagnetic/acoustic band structures and thus opens up an entirely new chapter in the realization of a plethora of emergent nonlinear physical phenomena. Research on nonlinear physics of moiré superlattices has gained remarkable interest and witnessed tremendous progress over the past few years. In this section, we briefly review the state-of-the-art nonlinear physics of moiré materials and structures, including but not limited to nonlinear electronics, optics and optoelectronics.

Nonlinear electronics of moiré superlattices

The Hall effect, one of the most representative electronic phenomena, which produces a voltage drop transverse to the direction of applied current, forms the basis of a wide range of topics in both

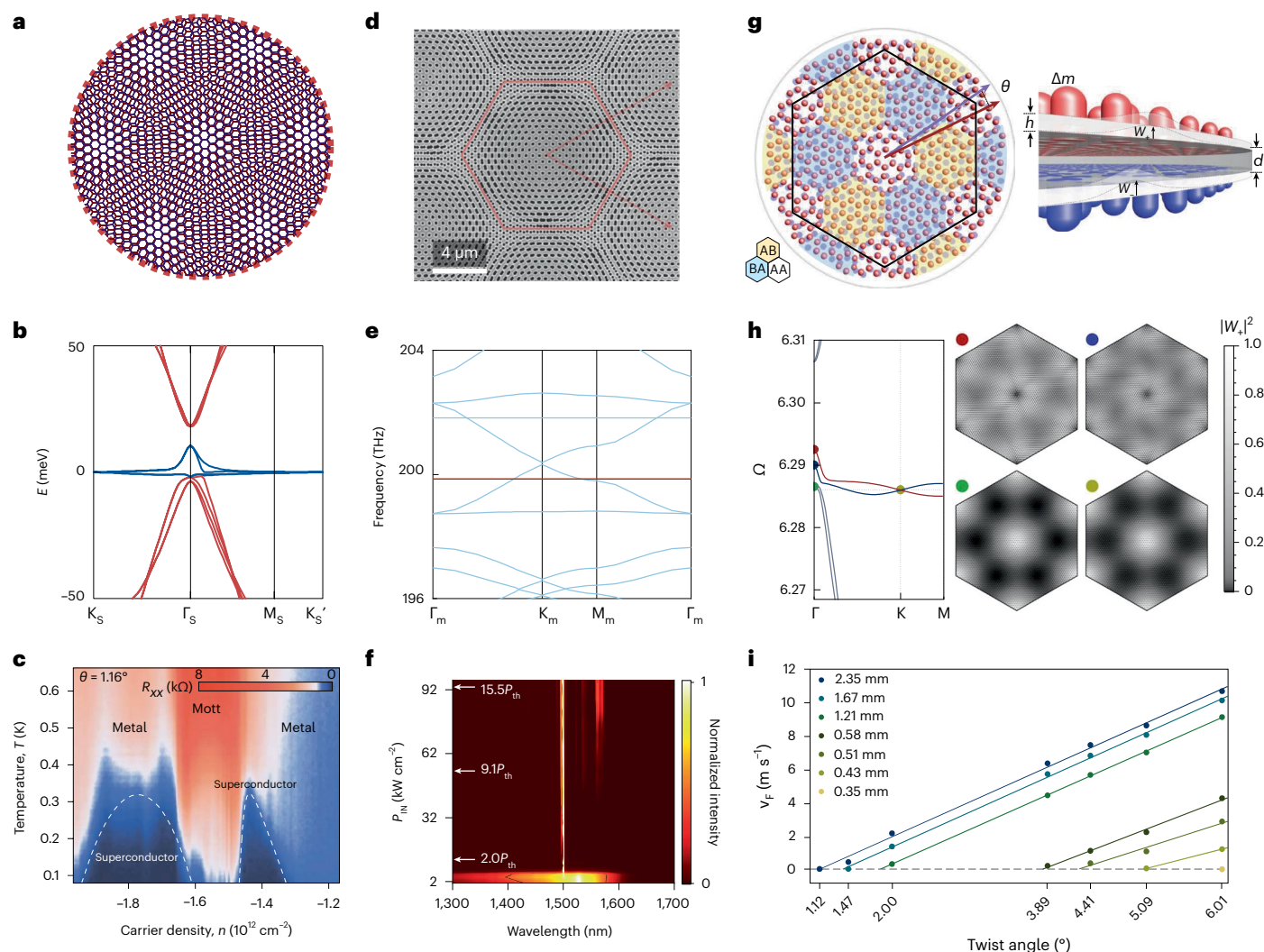


Fig. 2 | Moiré materials and structures. **a**, Schematic illustration of the TBG moiré superlattice. **b**, The band structure of magic-angle TBG ($\theta = 1.08^\circ$) calculated using an ab initio tight-binding method. The flat moiré bands are marked in blue. **c**, The phase diagram of magic-angle TBG ($\theta = 1.16^\circ$) against carrier density and temperature, showing superconducting domes, a correlated insulator phase and metallic states. **d**, Scanning electron microscopy image of a moiré structure formed by two twisted sets of photonic graphene lattices. The red hexagon (arrows) indicates the unit cell (lattice vectors) of the moiré superlattice. **e**, The calculated band diagram of a moiré photonic structure with a twist angle of 2.65° , showing flat moiré bands (red). **f**, Normalized spectra of the moiré structure shown in **d** for varying pump intensity (P_{in}), presenting a clear phase transition from broadband spontaneous emission to narrow lasing emission of

the moiré flat band mode at a threshold (P_{th}). **g**, Top (left) and side views (right) of a moiré phononic structure made of two vibrating plates patterned with a honeycomb mesh of masses and coupled across a continuum elastic medium. **h**, Low-energy (frequency) bands and spatial profile of selected eigenmodes (coloured dots) for a moiré phononic structure with a twist angle of 1.6° , showing the formation of flat bands and real-space localization of mechanical waves. **i**, The calculated Fermi velocity v_F of moiré photonic structures as a function of twist angle for different interlayer separations. For a certain separation, there is a typical twist angle (dubbed the magic angle) where v_F vanishes. Panels adapted with permission from: **a, b**, ref. 7, Springer Nature Ltd; **c**, ref. 6, Springer Nature Ltd; **e, f**, ref. 20, Springer Nature Ltd; **h**, ref. 37, APS; **i**, ref. 40, IOP Publishing. Panels reproduced with permission from: **d**, ref. 20, Springer Nature Ltd; **g**, ref. 37, APS.

fundamental research and practical applications (for example, the metrological resistance standard, magnetic field measurements and topological quantum computing)⁴⁵. Depending on whether the Hall response is at the same frequency as the driving electric field (of frequency ω_0), the Hall effects can be categorized into linear Hall effect (of frequency ω_0) and NHE (of frequency $m\omega_0$, integer $m \geq 2$). The linear Hall effect is constrained by time-reversal symmetry breaking and can manifest only in materials under an external magnetic field or magnets with a spontaneous internal magnetization. By contrast, the NHE (also referred to as nonlinear anomalous Hall effect) can go beyond this paradigm and occur in non-magnetic materials under zero magnetic field³.

According to Boltzmann transport theory, the conductivity tensor of the intrinsic second-order NHE can be written as

$\chi_{abc} = \varepsilon_{acd} D_{bd} \frac{e^3 \tau_r}{2\hbar^2 (1 + i\omega_0 \tau_r)}$, where e is the elementary charge, $\hbar$ denotes the reduced Planck constant, τ_r denotes the momentum relaxation time, ε_{acd} is the Levi-Civita symbol and D_{bd} represents the BCD, which is defined as the first-order moment of Berry curvature $\Omega(k)$ over the occupied states in momentum space, $D_{bd} = \int_k \frac{d^2 k}{(2\pi)^2} f_k \frac{\partial \Omega_d(k)}{\partial k_b}$, with $a, b, c, d \in \{x, y, z\}$ (Fig. 3a)^{3,46}. In contrast to the odd nature of $\Omega(k)$, $\frac{\partial \Omega_d(k)}{\partial k_b}$ is an even function with respect to the time-reversal symmetry. Consequently, a non-vanishing BCD and thus intrinsic second-order NHE can survive in time-reversal-invariant systems. On the other hand, a second-order NHE contributed from the BCD requires the breaking of both inversion and C_3 rotational symmetries according to symmetry analysis^{3,46}.

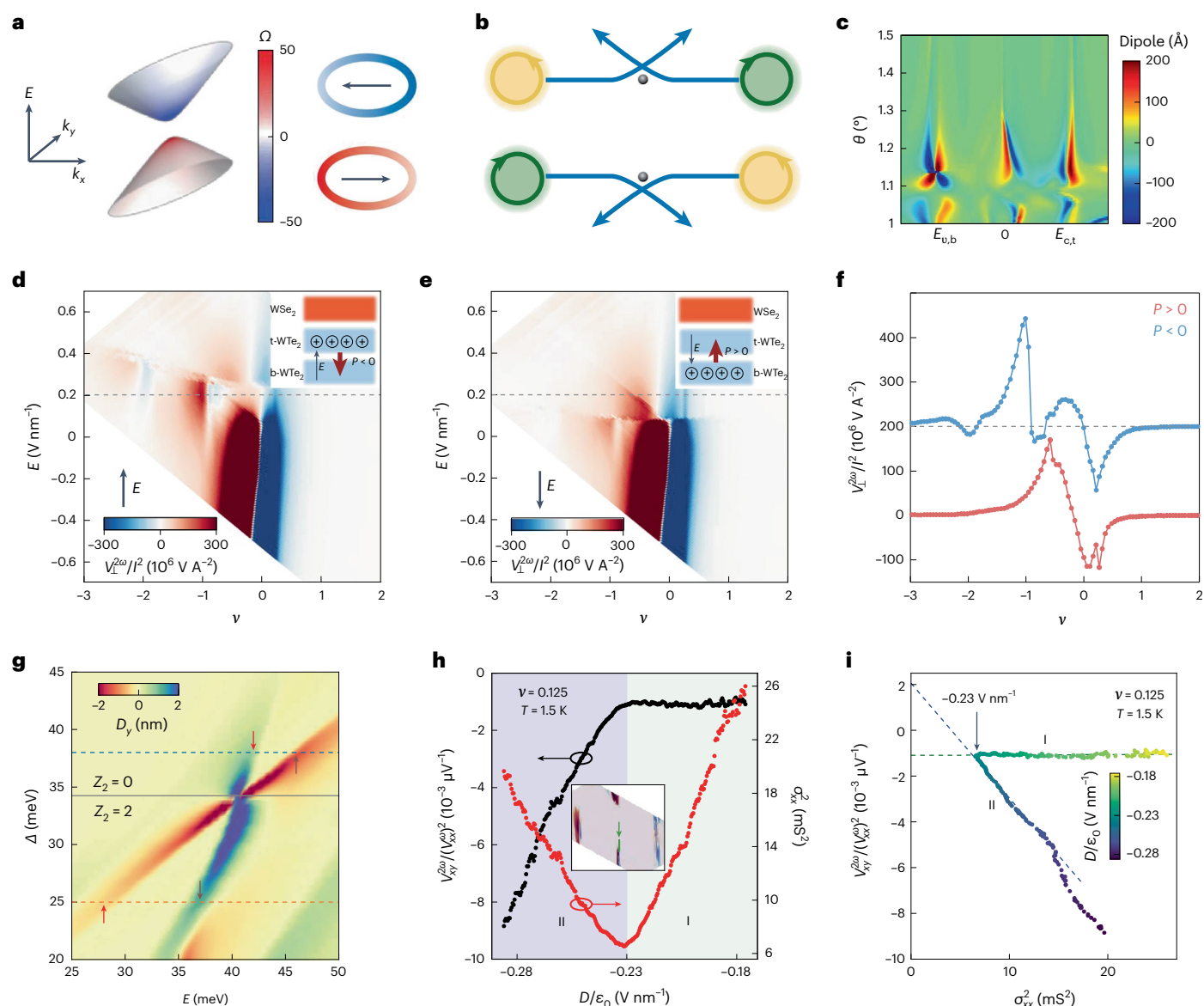


Fig. 3 | Nonlinear electronics of moiré superlattices. a, b, Schematic diagram of possible mechanisms of NHE: intrinsic BCD mechanism (**a**) and skew-scattering mechanism (**b**). **c**, Colour plot of calculated BCD of TBG as a function of θ and Fermi energy. A giant BCD of ~ 200 Å emerges near all band edges around the magic angle of 1.1° . $E_{v,b}$ ($E_{c,t}$) represents the energy level of the conduction (valence) band maximum (minimum). **d, e**, NHE response against the electric field and filling factor for the forward (**d**) and backward (**e**) electric field scan directions (black arrows). Inset: polarization-dependent hole (cross) distributions in ferroelectric bilayer WTe_2 . t- WTe_2 represents the top (bottom) WTe_2 layer. $V_L^{2\omega}/I^2$ denotes the NHE voltage response $V_L^{2\omega}$ normalized by the square of the driving current I . **f**, NHE response versus moiré filling factor for polarization $P < 0$ and $P > 0$ extracted from **d** and **e**, respectively, at $E = 0.2 \text{ V nm}^{-1}$

(black dashed lines in **d** and **e**). The horizontal dashed line marks the null NHE response for $P < 0$. The two curves are vertically displaced for clarity. **g**, Colour plot of BCD of TDBG as a function of Fermi energy E and interlayer potential Δ on the conduction-band side. The solid grey line denotes the Δ value across which the valley Chern number Z_2 of the flat conduction band changes with a sign reversal of the BCD. **h**, Normalized NHE responses (left axis) and square of longitudinal conductivity (right axis) of TDBG as a function of displacement field D at 1.5 K with $\nu = 0.125$. Inset: NHE responses versus ν and D . **i**, Scaling of NHE response with the square of longitudinal conductivity. Panels adapted with permission from: **a, b**, ref. 3, Springer Nature Ltd; **c**, ref. 47, APS; **d–f**, ref. 56, Springer Nature Ltd; **g–i**, ref. 52, Springer Nature Ltd.

Moiré materials offer an excellent platform for engineering NHEs by virtue of the non-trivial topological flat bands and various symmetry-breaking ground states induced by strong electron correlations (for example, nematic and stripe phases with spontaneous rotational symmetry breaking). Theoretically, giant BCD and intrinsic NHE have recently been predicted in strained TBG, TDBG and twisted bilayer TMD homobilayers^{47–49}. In particular, for a twist angle near the magic angle the BCD of weakly strained TBG can reach ~ 200 Å (Fig. 3c), two orders of magnitude larger than the state-of-the-art values reported in non-moiré materials⁴⁷. It is noteworthy that

the perfect twisted graphene or TMD moiré superlattices are not expected to exhibit an intrinsic NHE due to C_3 rotational symmetry. A small strain plays a crucial role in producing non-vanishing BCD and intrinsic NHE by breaking the rotational symmetry. Given the ubiquity of strain^{28,29,50}, strained moiré materials are actually a system closer to realistic circumstances. Experimentally, giant NHEs with intrinsic BCD contributions have been recently demonstrated in TBG⁵¹, TDBG^{52,53} and twisted bilayer MoS_2 ⁵⁴ and WSe_2 ⁵⁵. Notably, the NHE responses in twisted bilayer WSe_2 are enhanced by three orders of magnitude near the half-filling of moiré flat bands, evidencing

strong electronic correlations as a new mechanism for engineering nonlinear electronics⁵⁵.

Strikingly, in a moiré heterostructure consisting of a ferroelectric WTe₂ bilayer and a semiconductor WSe₂ monolayer, the NHE can be switched electrically (Fig. 3d–f)⁵⁶. A strong NHE response is observed near moiré band filling factors $\nu = -1, -2$ for electric field $E \leq 0.3 \text{ V nm}^{-1}$ in the forward scan (Fig. 3d) while only for $E \leq 0.1 \text{ V nm}^{-1}$ in the backward scan (Fig. 3e)⁵⁶. Such electrically switchable NHEs can be understood in terms of ferroelectric control of the moiré superlattice potential. Because the top WTe₂ layer interfaces with WSe₂ directly, it experiences a stronger moiré potential than does the bottom layer. For the forward scan, a negative polarization ($P < 0$) is prepared initially, and the mobile holes in bilayer WTe₂ are pushed to the top layer (inset of Fig. 3d). The strong moiré potential of the top layer holes drives the formation of Hubbard bands, resulting in enhanced BCD effects and NHE response. By contrast, a positive polarization ($P > 0$) is prepared first for the backward scan. Consequently, the mobile holes are localized at the bottom layer with negligible moiré superlattice potential (inset of Fig. 3e), leading to the absence of correlation effects and NHE responses⁵⁶.

Intriguingly, intrinsic NHEs introduce unique opportunities to underpin the topological phase transitions of moiré quantum materials, as the BCD is substantially enhanced near the topological phase transition point and changes sign when crossing it (Fig. 3g)^{48,49,52}. It is noteworthy that probing topological phase transitions by linear means is extremely challenging due to the absence of local order parameters⁵². By employing NHE measurements and scaling analysis (Fig. 3h,i), a topological phase transition of valley Chern type driven by out-of-plane electric fields has been successfully unravelled in TDBG⁵². Additionally, we remark that, apart from the intrinsic BCD mechanisms, the NHE can also originate from quantum metric dipole⁵⁷ and extrinsic contributions, such as side-jump mechanisms and skew scattering (Fig. 3b)³. Recently, an NHE induced by the quantum metric dipole has been reported in a topological antiferromagnetic heterostructure where the crystalline a axes of the PT -symmetric MnBi₂Te₄ (P , parity inversion; T , time reversal) and black phosphorus (BP) are aligned⁵⁷. We highlight that this is a landmark advance in condensed-matter physics given that the effects of the quantum metric have been long sought but rarely explored. In addition, extrinsic NHEs from skew scattering of chiral Bloch electrons have been uncovered in TBG, TDBG and graphene/h-BN moiré superlattices^{53,58,59}. Remarkably, an extrinsic NHE with extremely high conductivity of $\sim 500 \mu\text{m S V}^{-1}$ is detected near the van Hove singularities in TDBG, which is orders of magnitude higher than the best values reported in non-moiré materials ($\sim 1 \mu\text{m S V}^{-1}$)⁵³. Such a giant moiré NHE response, which is expected to be further enhanced through external means (such as ultrafast optical excitations, strain, magnetic field and pressure), would underlie the development of a large portfolio of emerging applications, such as frequency doubling, wireless communication and terahertz detection.

Nonlinear photonics of moiré superlattices

Nonlinear optics, a branch of modern optics that deals with light–matter interactions under high-intensity optical fields, is fundamentally important to both basic scientific studies (such as spin-order probing, ferroelectricity detection and attosecond physics) and technological innovations (for example, entangled photon pair generation, electro-optic modulators, all-optical signal processing and ultrashort coherent pulse generation)^{1,2}. Recent studies show that moiré superlattices have emerged as an exciting platform for nonlinear optics and quantum photonic technologies. In this section, we overview the state-of-the-art nonlinear photonics of moiré superlattices, such as SHG, THG, HHG, optical nonlinearity of quasiparticles and optical solitons.

Optical harmonic generation. SHG is one of the most widely studied nonlinear optical harmonic generation processes, in which new

photons with twice the energy of incident photons are generated. According to the nonlinear optical principles, SHG within the electric-dipole limit is largely constrained by crystalline symmetries and can manifest exclusively under conditions of broken inversion symmetry¹. Specifically, moiré superlattices with twist-angle-dependent inversion symmetry breaking afford extraordinary opportunities for dynamical control of the SHG responses. By dynamically rotating the twist angles with the nanomechanical technique based on atomic force microscope tip manipulation (Fig. 4a)⁶⁰, continuously tunable SHG responses are observed in twisted h-BN moiré interfaces (Fig. 4b)⁶¹. When the twisted h-BN interfacial configuration possesses the largest degree of broken inversion symmetry, the SHG intensity is the brightest. By contrast, the SHG response almost disappears when the interfacial configuration is centrosymmetric. Furthermore, by engineering phase-matching with twist angle, an extremely high SHG conversion efficiency of $\sim 8\%$ and a facile polarization controllability unavailable in conventional crystals have been achieved in twisted rhombohedron h-BN superlattices⁶². Such a dynamically twistable SHG with extremely tunable nonlinear efficiency and twist-phase-matching engineering would hold great promise for a wide variety of cutting-edge applications, such as compact optomechanical transducers, ultrafast photonics, optical information processing, tunable light sources, biosensing and quantum optics.

Aside from the twisted h-BN moiré interfaces, SHG efficiency that is dynamically controllable with twist angle has also been uncovered in TBG and twisted TMD superlattices^{63–66}. More recently, it has been demonstrated that twisted moiré materials enable control over the intrinsic point group symmetries and thus previously forbidden new SHG responses⁶⁷. In contrast to monolayer TMDs with only an in-plane susceptibility element $\chi_{yyy}^{(2)}$, twisted TMD bilayers at a generic twist angle ($\theta \neq 0^\circ$ or 180°) can harbour the new chiral element $\chi_{xyz}^{(2)}$, because the point group is reduced from monolayer D_{3h} to twisted D_3 (Fig. 4c)⁶⁷. Interestingly, the new chiral SHG response is independent on the in-plane sample orientation and therefore can coherently add up when scaling up to bulk while keeping the twist angle invariant, promising new nonlinear functionalities for a plethora of fascinating phenomena and applications. In addition, theoretical calculations predict that, because all electronic states share the same resonant frequency, moiré flat bands with van Hove singularities in the density of states would significantly enhance the resonant effect and therefore give rise to colossal SHG responses at extremely low frequencies, such as the terahertz and far-infrared frequency regime^{68,69}. For example, the calculated SHG susceptibilities of h-BN aligned TBG moiré superlattices can be up to 10^6 pm V^{-1} at low frequencies $\hbar\omega \leq 0.1 \text{ eV}$, orders of magnitude larger than those previously reported in the best nonlinear optical materials⁶⁸. Detecting such terahertz moiré SHG with giant nonlinearity can promise a broad range of emerging applications, albeit technically challenging owing to the lack of frequency-selective components, intense light sources and sensitive detectors in the terahertz regime⁴.

In addition to SHG, twist angle also opens a new structure degree of freedom to engineer the THG susceptibility. For instance, the THG efficiency of TBG is strongly increased near a critical twist angle θ_c , where the transition energies between the van Hove singularities of conduction and valence bands fulfil the three-photon resonance of incident light⁷⁰. More recently, emerging THG circular dichroism with near-unity polarization, which cannot be attained in constituent monolayers, has been reported in twisted four- and eight-layer TMD structures with newly added screw symmetry (Fig. 4d)⁶⁷. The giant nonlinear optical chirality enabled by twist-angle engineering offers extraordinary opportunities for spin–orbit coupling of light and innovative chiroptical applications in nanophotonics and optoelectronics⁷¹. In addition, moiré superlattices would have a strong impact on HHG and other nonlinear effects (such as two-photon photoluminescent and four-wave mixing), given that these processes are extremely sensitive to symmetry breaking and electronic band structure^{72,73}. Indeed, a

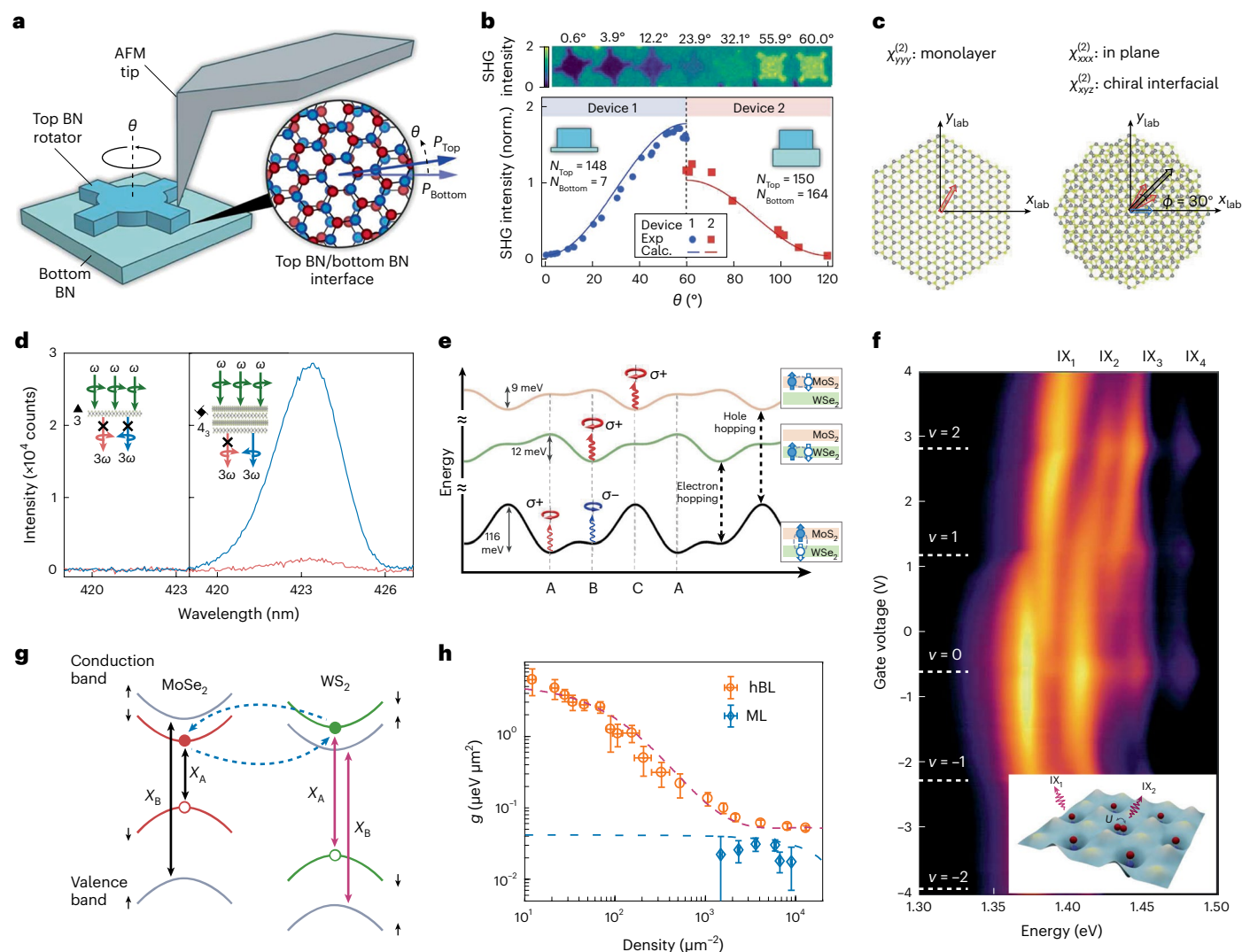


Fig. 4 | Nonlinear photonics of moiré materials. **a**, Schematic of dynamical tuning of twist angle based on an atomic force microscope tip manipulation technique. Inset: top view of the twisted h-BN moiré interfaces. **b**, Bottom: SHG intensity of h-BN moiré interfaces versus θ . Top: SHG mapping for some specific θ . **c**, Atomic structures and SHG polarizations in a WS_2 monolayer (left) and 30° twisted bilayer (right). **d**, Helicity-resolved THG for monolayer (left) and twisted four-layer WS_2 (right). **e**, Moiré potential landscapes for the intra- and interlayer excitons in an angle-aligned $\text{MoSe}_2/\text{WSe}_2$ heterostructure. The excitons are trapped at the moiré potential minima. **f**, Interlayer exciton PL intensity against doping and photon energy at high exciton density in a twisted WS_2/WSe_2 heterostructure. IX₁ denotes the moiré trapped single-occupancy state, IX₂–IX₄

are the multi-occupancy states. Inset: schematic diagram showing the moiré trapped single- (IX₁) and double-occupancy states (IX₂–IX₄). **g**, Schematic diagram of resonantly hybridized excitons of angle-aligned $\text{MoSe}_2/\text{WSe}_2$ heterostructure resulting from interlayer electron hopping. **h**, Nonlinear coefficient of polaritons g as a function of carrier density for $\text{WS}_2/\text{MoSe}_2$ moiré heterobilayer (hBL) and monolayer MoSe_2 (ML). Panels adapted with permission from: **a, b**, ref. 61 under a Creative Commons license CC BY-NC 4.0; **c, d**, ref. 67, Springer Nature Ltd; **e**, ref. 75 under a Creative Commons license CC BY-NC 4.0; **f**, ref. 77, Springer Nature Ltd; **h**, ref. 80, Springer Nature Ltd. Panels reproduced with permission from: **f**, inset, ref. 77, Springer Nature Ltd; **g**, ref. 79, Springer Nature Ltd.

theoretical calculation shows that TBG can exhibit enriched even-order HHG that cannot occur in monolayer or conventional bilayer graphene⁷⁴. The exploration of optical harmonic generation in moiré superlattices has just started, and the future progression will certainly deepen our understanding of nonlinear physics and stimulate technological advances in novel photonics, electronics and twistronics.

Optical nonlinearity of quasiparticles. Moiré superlattices impose a new periodic potential landscape on elementary excitations, for example, excitons. When the moiré period is larger than the exciton Bohr radius, excitons would be trapped at the moiré potential minima, producing an ordered arrangement of laterally localized quantum-dot-like excitons (Fig. 4e)^{8,17,75}. For moiré-trapped excitons, especially for interlayer ones with non-vanishing out-of-plane static electric dipoles, the dipole–dipole interactions would result in an

on-site photon repulsion, and therefore induce effective photon–photon coupling and optical nonlinearity⁷⁶. Recently, strong optical nonlinearity of moiré-localized interlayer excitons, evidenced as an exciton-density-dependent blueshift of the emission energy and the emergence of high-energy multi-occupancy states at high exciton density, has been demonstrated in WS_2/WSe_2 and $\text{MoSe}_2/\text{WSe}_2$ heterostructures (Fig. 4f)^{76–78}. The giant nonlinearity suggests that moiré-trapped interlayer excitons provide a potential platform for many-body exciton physics, mixed Bose–Fermi Hubbard model and efficient photon switching applications.

Resonantly hybridized excitons, a hybrid quasiparticle excitation of intralayer excitons and interlayer excitons, emerge in moiré superlattices with the alignment of band edges (such as $\text{MoSe}_2/\text{WSe}_2$ moiré heterostructures with nearly degenerate conduction-band edges, Fig. 4g)^{35,79–81}. Resonantly hybridized moiré excitons feature the

merits of both intralayer excitons (for example, strong interactions with light) and interlayer excitons (for example, appreciable static dipole moment), facilitating the realization of hybrid matter–light excited states (that is, exciton–polaritons) with strong optical nonlinearity^{80,82}. By integrating resonantly hybridized moiré excitons into a microcavity, moiré-induced nonlinearity of exciton–polaritons in the strongly coupled regime of light–matter interactions has been recently revealed in MoSe₂/WS₂ moiré superlattices⁸⁰. The effective polariton nonlinearity is enhanced by $(a_M/a_B)^2$, where a_M and a_B are the moiré period and exciton Bohr radius, respectively. The nonlinear coefficient of moiré exciton–polaritons can reach up to $-6 \mu\text{eV} \mu\text{m}^2$ (Fig. 4h), which is more than two orders of magnitude larger than results for monolayer TMDC exciton–polaritons^{80,83}. Further, a higher nonlinearity of moiré exciton–polaritons could be achievable by optimizing the size/depth of moiré potential, and/or by external control (for example, electric/magnetic fields and strain/pressure engineering).

Optical solitons. Optical solitons, a salient example of nonlinear electromagnetic responses, emerge as a result of the delicate balance between diffraction and self-phase modulation induced by nonlinearity, and hold great promise for unique technological advances in information transmission, all-optical routing, light-controlling-light and so on²⁴. Moiré photonic structures with flattened bands in their spectra are fertile ground for engineering optical solitons. Indeed, by combining experimental measurements with numerical simulations of the nonlinear Schrödinger-like equation, optical solitons are evidenced unequivocally in a reconfigurable moiré photonic superlattice optically induced in photorefractive media²⁴. Significantly, moiré optical solitons strongly depend on the twist angles and are intimately connected with the transition between commensurate and incommensurate geometries (that is, the periodic–aperiodic structural transition)²⁴. For most twist angles, moiré photonic structures are incommensurate (that is, an aperiodic geometry; lower panel of Fig. 5a). The incommensurate moiré structures show a localization–delocalization transition of light when the amplitudes of the constituent photonic sublattices are tuned to exceed a critical value^{23,24}. Above the threshold amplitude, some photonic modes are localized (Fig. 5b). Consequently, optical solitons can bifurcate from the respective linear localized mode and emerge in the absence of excitation power (that is, thresholdless, Fig. 5c–e)²⁴.

By contrast, when the twist angle is at certain specific values (such as the Pythagorean angles in the case of twisted bilayer square lattices) the moiré photonic structure is commensurate (upper panel, Fig. 5a)^{23,24}. Because all photonic eigenmodes are delocalized (Fig. 5b), the formation of optical solitons requires a certain excitation power to balance the diffraction (Fig. 5c–e). Interestingly, it was found that the threshold power of the optical soliton decreases continuously as the order of the Pythagorean angle increases (Fig. 5d). The underlying physics is that the photonic bands are progressively flattened with increasing Pythagorean angle order²⁴. Notably, for the third-order Pythagorean angle moiré superlattice, the threshold power of the optical soliton is extremely low, more than three orders of magnitude smaller than that in a simple square lattice under the same parameter conditions. The realization of long-sought optical solitons at ultralow power thresholds forms a solid foundation for real-world applications.

Besides the power in the investigation of Kerr-like optical solitons, moiré structures may also have important implications for topologically complex solitons, such as multifrequency solitons, out-of-phase dipole solitons, vortex solitons, corner solitons and edge solitons^{84,85}. In addition, moiré structures would promise the prospect of nonlinear optical enhancement. For example, by engineering the twist angle and interlayer coupling strength to form flat bands above the light line, a new optical state with dual merits of moiré flat bands and bound states in the continuum (BICs), dubbed a moiré quasi-BIC, is constructed in

twisted photonic superlattices (Fig. 5f)⁸⁶. The SHG efficiency assisted by this moiré quasi-BIC is enhanced by nearly an order of magnitude, compared with that based on conventional quasi-BICs with a similar Q factor (Fig. 5g)⁸⁶. By designing a photonic moiré superlattice with two synthetic twist angles (that is, a moiré synthetic sphere, Fig. 5h), transition photonic bands with optimal band compression are obtained at a series of magic-angle pairs⁸⁷. The band compression in a synthetic moiré sphere can lead to field enhancement and thus enhance the nonlinear optical effect. For instance, the SHG generated in the moiré synthetic sphere is nearly 50 times larger than that produced in the substrate (Fig. 5i)⁸⁷.

Further, moiré metasurfaces, composed of quasiperiodic structures with slight displacements, offer advantages in precise control and dynamic tunability of structural configurations, enabling unique electromagnetic functionalities—for example, topological transitions⁸⁸. Recently, time-modulated metasurfaces have shown great promise in manipulating microwave harmonic waves by simultaneously controlling space and frequency domains. Extending time modulation to synthetic moiré structures can present new opportunities (for example, dynamic beamforming) for engineering nonlinear electromagnetic waves⁸⁹. We highlight that we are just starting down the path of exploring the nonlinear optics of moiré structures, and more new nonlinear phenomena will emerge in the future, such as nonlinear optical chirality, nonlinear geometric Berry phase, multistability and nonlinear optical wavefront engineering.

Nonlinear optoelectronics of moiré superlattices

Optoelectronics is a key branch of modern physics concerned with electronic devices for the generation, detection and control of light, such as photovoltaics, electro-optic modulators and light-emitting diodes. Current optoelectronics operates mainly in the linear-response regime. A notable manifestation of nonlinear optoelectronics is the BPGE, intricately connected with the quantum geometry and topology of electron wavefunctions⁴. Benefiting from its nonlinear nature, BPGE can circumvent the thermodynamic Shockley–Queisser limit of the traditional linear photovoltaic effect, and therefore lays a firm foundation for a wide range of innovative technologies, including clean energy, all-optical transistors, imaging and chemical and biological sensing^{4,5}. Typically, BPGE can originate from two contributions: shift current associated with the real-space displacement of an electron wave packet upon interband photoexcitation, and injection current arising from the change in group velocity of photoexcited electrons and holes^{4,5}. Note that these two contributions are constrained by the spacetime symmetries of crystals. In the presence of time-reversal symmetry, the contribution of shift current is observed with linearly polarized light excitations and corresponds to the LPGE, whereas the contribution of injection current occurs under circularly polarized irradiation and refers to the CPGE^{4,5}.

Moiré superlattices with electron–correlation–boosted non-trivial quantum geometric and topological properties introduce an emergent platform for tunable BPGE applications. Theoretically, giant CPGE and LPGE responses attaining magnitudes stronger than previously seen have recently been uncovered in TBG moiré superlattices (Fig. 6a)^{90–92}. In particular, the pronounced CPGE and LPGE responses of TBG are situated in the long-sought terahertz range, holding great promise for overcoming the so-called terahertz gap in the field of terahertz technology⁹³. Experimentally, exotic CPGE and LPGE in the terahertz region have been demonstrated in TBG close to the second magic angle of -0.6° (Fig. 6b)^{94,95}. Intriguingly, the CPGE and LPGE responses display pronounced oscillations upon variation of the gate voltage (right panel, Fig. 6b), which can be understood as the shifting of the Fermi energy across the multitude of moiré bands in TBG^{94,95}.

As well as the terahertz BPGE in TBG, strong mid-infrared CPGE and LPGE at $\sim 5 \mu\text{m}$ and $7.7 \mu\text{m}$, originating from the moiré-engineered symmetry breaking and quantum geometric properties of Bloch

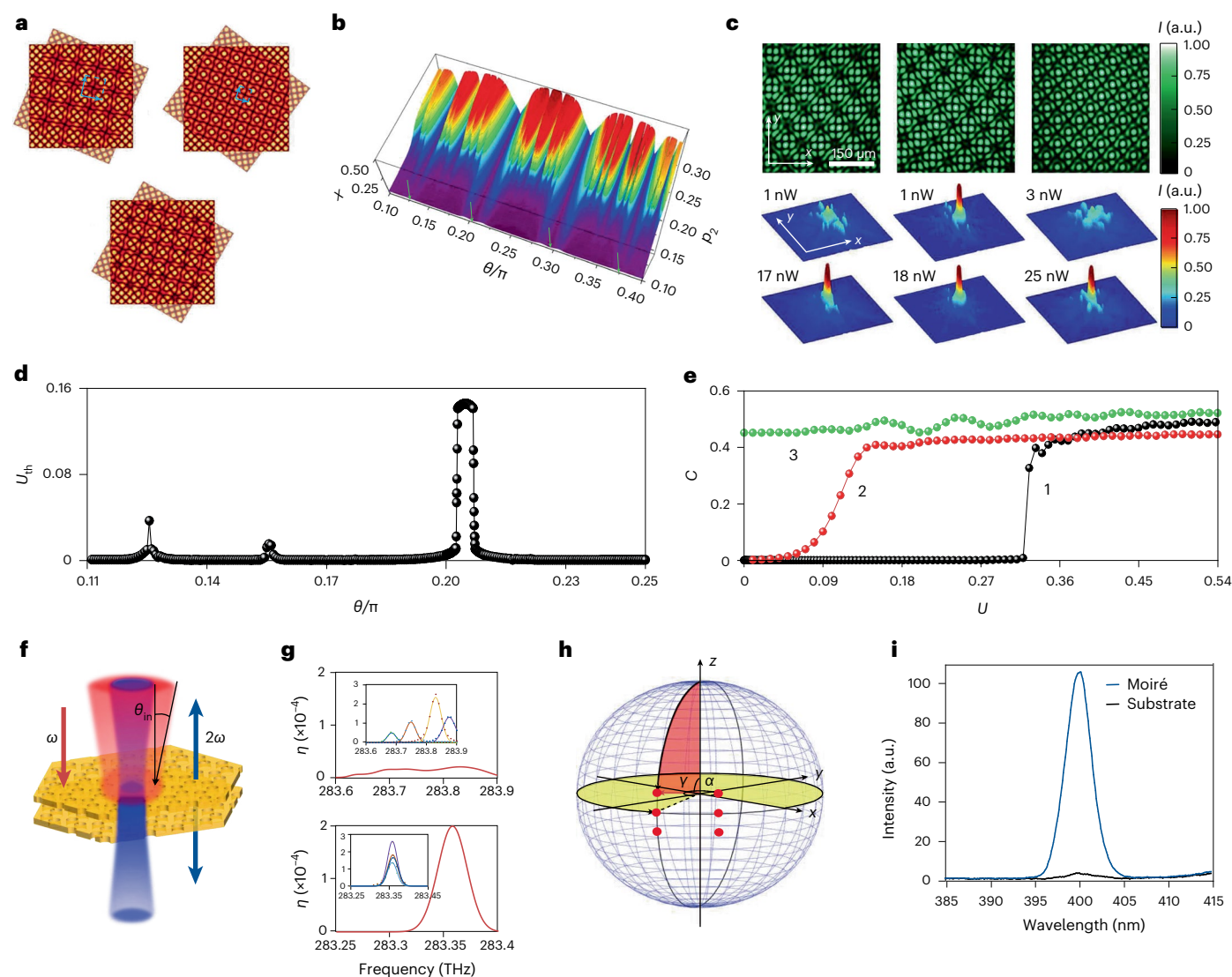


Fig. 5 | Nonlinear optics of moiré structures. **a**, Top: commensurate moiré photonic lattices formed by two square sublattices with Pythagorean angles $\theta = \arctan(5/12)$ (left) and $\theta = \arctan(3/4)$ (right). Blue arrows are the primitive lattice vectors. Bottom: incommensurate moiré photonic lattices with $\theta \approx 0.167\pi$. **b**, Form factor (the degree of mode localization) against θ and the amplitude p_2 of the second square sublattice. The amplitude of the first square sublattice is fixed at $p_1 = 0.5$. Green arrows indicate Pythagorean angles. **c**, Lattice profiles (top) and observed output signal intensity distribution corresponding to low (middle) and high (bottom) excitation powers, for $\theta = \arctan(5/12)$ (left), $\theta = 0.139\pi$ (middle) and $\theta = \arctan(3/4)$ (right). **d**, Thresholds of soliton formation as a function of θ . **e**, Optical soliton content against excitation power

for Pythagorean angles $\theta = \arctan(3/4)$ (curve 1) and $\theta = \arctan(5/12)$ (curve 2) and for non-Pythagorean angle $\theta \approx 0.139\pi$ (curve 3). For incommensurate moiré photonic lattices, the excitation of the soliton requires no power, indicating that it is thresholdless (curve 3). **f**, Schematic illustration of SHG enhanced by moiré quasi-BICs under a wide-angle source. **g**, Efficiency of SHG assisted by moiré quasi-BICs (bottom) and based on dispersive quasi-BICs with similar Q factors (top). **h**, Schematic of synthetic moiré sphere defined with two synthetic angles (α, γ). **i**, Measured SHG intensity of synthetic moiré sphere and substrate. Panels adapted with permission from: **a, c–e**, ref. 24, Springer Nature Ltd; **g**, ref. 86, APS; **h, i**, ref. 87, APS. Panels reproduced with permission from: **b**, ref. 24, Springer Nature Ltd; **f**, ref. 86, APS.

wavefunctions, have been recently evidenced in TDBG moiré superlattices (Fig. 6c)²⁵. Remarkably, because of the electrical tunability of moiré quantum geometry (for example, interband Berry connections and interband BCD), the LPGE and CPGE responses in TDBG superlattices are highly tunable with the out-of-plane displacement field and doping density. For example, by changing the gate voltage biases, the magnitudes of the CPGE photovoltages can be adjusted by a factor of ~ 5 , whereas the polarities can be switched by 180° (Fig. 6c)²⁵. Benefiting from the strong, electrically tunable BPGE in TDBG moiré superlattices, intelligent infrared sensing with a subwavelength footprint of merely $\sim 3 \times 3 \mu\text{m}^2$ has been successfully constructed, which is capable of inferring the fingerprint of light, for example, polarization, power and wavelength²⁵.

Notably, moiré superlattices formed by stacking crystals with different crystalline symmetries offer extraordinary opportunities to engineer the interfacial symmetry breaking and, therefore, exotic BPGE responses. For example, by vertically stacking WSe_2 and BP/silicon phosphide (SiP) with distinct rotational and mirror symmetries (that is, WSe_2 has C_3 rotational symmetry and mirror lines exist along the armchair direction, while BP and SiP display C_2 rotational symmetry and sets of mirror planes), a stripe moiré superlattice with only a single mirror line is formed (Fig. 6d)^{96,97}. By virtue of the strong symmetry reduction, a non-vanishing BCD perpendicular to the mirror line emerges in WSe_2/BP and WSe_2/SiP moiré superlattices with a twist angle of 0° , resulting in the spontaneous quantum-mechanical shift/injection currents and nonlinear quantum response of BPGE

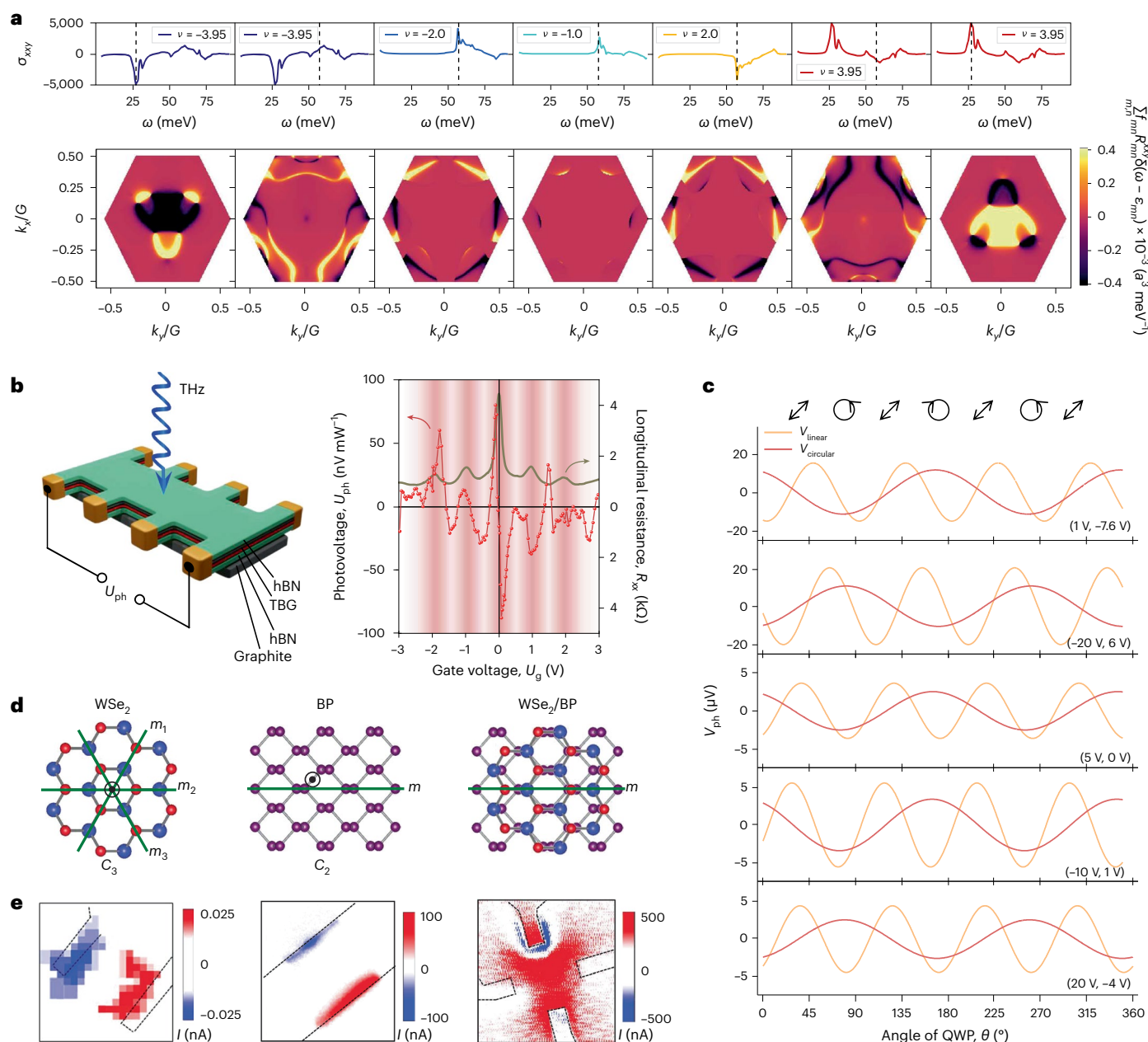


Fig. 6 | Nonlinear optoelectronics of moiré superlattices. **a**, Top: calculated shift-current conductivity as a function of photon frequency ω at some specific ν . Bottom: k -space profile of the calculated shift-current conductivity at the frequency corresponding to the dashed line in the upper panel. **b**, Left: schematic of a TBG device excited by a terahertz laser for BPGE measurement. Right: BPGE responses (red) and the longitudinal resistance (grey) as a function of gate voltage for TBG twist close to the second magic angle. **c**, CPGE and LPGE

responses against the angle of the quarter-wave plate (QWP) at different gate voltages. **d**, **e**, Schematic illustrations (**d**) and photocurrent mappings (**e**) of monolayer WSe_2 , BP and WSe_2/BP moiré superlattice. Green lines and circled dots represent mirror planes and rotational axes, respectively. Panels adapted with permission from: **a**, ref. 90 under a Creative Commons license CC BY 4.0; **b**, ref. 94, American Chemical Society; **c**, ref. 25, Springer Nature Ltd; **e**, ref. 96, AAAS. Panel **d** reproduced with permission from ref. 96, AAAS.

(right panel, Fig. 6e)^{96,97}. It is noteworthy that the exotic nonlinear optoelectronic phenomena are originally prohibited in the constituent WSe_2 , BP and SiP because of the symmetry constraints. Although photocurrent responses can also occur around the electrodes of the constituent monolayer WSe_2 and BP/SiP (left and middle panels of Fig. 6e), they are due to the contact-induced photothermal/Schottky barrier effects in the linear regime, rather than the intrinsic nonlinear BPGE mechanism, which shows the largest response when the incident light illuminates the centre of the devices (Fig. 6e)^{96–98}. Further, strong nonlinear optoelectronic responses and device applications, particularly in the fascinating terahertz range, are expected to be

demonstrated in a wide variety of other moiré superlattices, such as magic-angle TBG, rotationally aligned graphene/h-BN, twisted bilayer WSe_2 , $\text{WTe}_2/\text{WSe}_2$ and topological antiferromagnetic $\text{MnBi}_2\text{Te}_4/\text{BP}$ superlattices, in which giant nonlinear electric transport responses under low driving frequencies have been uncovered (see ‘Nonlinear electronics of moiré superlattices’)^{54–56,58,59}.

Perspectives

In the span of less than five years, progress in nonlinear physics of moiré superlattices has grown substantially and seen a delightful series of discoveries, such as the giant quantum NHE, dynamic harmonic

generation, thresholdless optical solitons, strong optical nonlinearity of elementary excitations, electrically tunable CPGE/LPGE and so on. Building on these early successes and discoveries announced almost every week, we highlight that this is just the beginning of nonlinear physics of moiré materials and structures, and there are plenty of new opportunities and emerging technological applications to be eagerly awaited.

New nonlinear physics and engineering

Given our exploratory foray into the nonlinear physics of moiré superlattices, one intriguing research frontier is to explore new nonlinear physical phenomena in moiré materials and structures, such as the nonlinear spin Hall effect, gyrotropic Hall effect, non-reciprocal thermoelectric transport, stimulated Raman scattering, optical parametric oscillators, spontaneous parametric downconversion, saturable absorption, nonlinear optical phase engineering, optical bistability and switching. In particular, the current research mainly focuses on nonlinear physics of moiré superlattices in the perturbative regime. Considering the strongly enhanced nonlinear responses, moiré superlattices promise the prospect of advancing nonlinear physics to the non-perturbative or extreme nonlinear regime. It is noteworthy that non-perturbative, extremely high-order nonlinear effects in condensed matter are a long-sought goal of modern physics and materials science, and have important implications for attosecond science, strong-field physics and non-equilibrium band-structure-related phenomena, as well as a plethora of emergent technological applications^{72,99}. Limited by the relatively low conversion efficiencies, the current state-of-the-art order of extreme nonlinear effects in non-moiré solids is typically less than 50th^{72,99}. Moiré superlattices offer a powerful solid-state quantum platform to address the bottlenecks in non-moiré crystals and greatly facilitate the extraordinary opportunities of non-perturbative nonlinear physics up to hundreds of orders and beyond.

Meanwhile, nonlinear responses afford unparalleled possibilities to create, probe and manipulate a rich phase diagram of quantum phases that are unattainable within the framework of linear approaches. For instance, topological phase transitions and hidden orders that are invisible to conventional linear probes, such as non-dipolar magnetic order in antiferromagnetic Mott insulator Sr_2IrO_4 and multipolar electronic nematic order in spin-orbit-coupled metal $\text{Cd}_2\text{Re}_2\text{O}_7$, have been unravelled successfully by taking advantage of nonlinear techniques^{5,52,100,101}. Consequently, it would be highly desirable to employ nonlinear effects as a versatile and multiphysics diagnostic tool for the emerging moiré quantum phases. By employing angle-resolved NHE measurements, a new type of cascade phenomenon and new insights into the nature of spontaneous symmetry breaking have been revealed in a twisted trilayer graphene moiré superlattice¹⁰². We also highlight that the engineering of quantum moiré superlattices with nonlinear techniques (for example, Floquet engineering with time-periodic light field, chiral optical structures, ultrafast and ultrastrong light fields) will create exotic non-equilibrium physics and new functionalities, such as non-Abelian excitations, photon-driven anomalous Hall effect, light-mediated superconductivity, ferroelectricity, magnetism, topology and dynamical gap opening^{103–106}. Further, controlling emergent moiré phenomena with nonlinear effects will certainly advance our understanding of the underlying mechanism and greatly facilitate the further development of next-generation applications in nonlinear electronics, optics and optoelectronics, such as ultracompact quantum light sources, high-performance photon modulators, intelligent infrared sensors, wireless communications and on-chip terahertz wave technology.

Moiré superlattice design and engineering

Another future prospect is to design new moiré superlattices with on-demand nonlinear properties. Here, we present some prototype moiré superlattices with great potential to outline a bright vision for

nonlinear physics. First, chiral topological Weyl lattices without inversion and mirror symmetries are predicted to show fascinating nonlinear responses beyond those in non-chiral semimetals, such as quantized CPGE in terms of the fundamental constants (that is, $C \frac{e^3}{4\pi\hbar^2}$, where C is the topological charge of a monopole)^{4,107}. However, the scarcity of chiral Weyl semimetals has greatly hindered the progress of this field. Moiré superlattices formed by Weyl electronic, photonic or phononic crystals are promising in this regard and might open up new paradigms for chiral Weyl lattices and the long-awaited goal of quantized topological nonlinear responses. Second, magnetic moiré superlattices, especially those composed of A-type antiferromagnetic crystals (for example, CrI_3 , CrSBr and MnBi_2Te_4), facilitate control of magnetic order, interlayer electronic coupling, exchange interactions and topology in a flexible fashion^{108–110}. This promises a large class of programmable nonlinear effects for future applications, such as non-reciprocal SHG and spin BPGE^{111,112}. Third, for moiré superlattices made of two materials with distinct symmetry-breaking orders (for example, ferroelectric crystals with inversion asymmetry and anisotropic materials with broken rotational symmetry), the interplay between the breaking of different symmetries could result in synergistic effects with $1 + 1 > 2$ and provide extraordinary opportunities to underpin completely new nonlinear physics and device applications⁵⁰. Fourth, moiré superlattices composed of strongly correlated or topological phases, such as superconductors (for example, NbSe_2), quantum spin liquids (for example, RuCl_3) and charge density waves (for example, TaS_2), could also offer promising avenues for a large portfolio of tantalizing nonlinear physical phenomena. Last but not least, moiré superlattices, by engineering through external means (for example, electric/magnetic fields, strains and pressure) or by integrating with other systems (for example, cavities, plasmonics, waveguides and integrated optical circuits), could show strongly enhanced nonlinearity and potentially possess a wide range of applications in on-chip ultrafast nanophotonics, attosecond pulse generation, metrology, biomicroscopy, compact terahertz sources and quantum optical technologies.

Dynamic moiré nonlinear superlattices

Research on artificial moiré superlattices is still in its infancy, with many new developments anticipated. One key research direction is to demonstrate the possible existence of topologically complex states in artificial moiré superlattices (for example, vortex solitons with an intrinsic orbital angular momentum). Due to the susceptibility of vortex solitons to destabilization through the azimuthal instability, it is intriguing to examine how inhomogeneous refractive index distributions in moiré structures with varying geometries and twisting angles impact the azimuthal instability of vortices. For example, it is possible that vortex solitons could preserve their stability over long distances under lower-power conditions. Thus, it is natural to expect forthcoming theoretical and experimental studies of moiré vortex solitons, and possible self-trapping mechanisms involving the parametric wave-mixing process, such as sum- and difference-frequency generation and four-wave mixing.

Another unique aspect of moiré structures, inaccessible with other physical systems, is their ability to vary in dimensionality. While the current investigations are limited to two dimensions (that is, stationary), exploring the possibility of upgrading to three dimensions (that is, dynamic) offers an exciting direction. These dynamic moiré nonlinear structures provide all-space localization of different waves (for example, optical, acoustic, mechanic, microwaves and even charge density waves) and hold great promise for emerging nonlinear photonic and optoelectronic applications, such as quantum optics, cavity quantum electrodynamics, nanolasers, nano light-emitting diodes, detectors, sensors and modulators. By introducing further modulation along the time dimension for the formation of time moiré lattices, unprecedented possibilities could emerge for steering three-dimensional

localized wave packets in different directions with controlled velocities and high nonlinearity. The integration of two-dimensional moiré structures with a third evolutionary dimension suggests that authentic three-dimensional moiré superlattices are on the horizon¹¹³, providing the potential for new functional devices, such as ultrafast switching and modulating devices operating in ultralow-power conditions for use in optical communication, computing and optical neural networks.

Challenges towards practical applications

Despite many promising aspects, there are still a number of grand challenges that remain to be overcome. One of the particularly critical challenges impeding the immediate progress towards real device applications in the nonlinear physics of moiré superlattices is the lack of reproducibility and scalability of the moiré materials and structures, because of the ubiquity of sizable twist-angle inhomogeneity and various disorders (for example, uncontrolled strain and interfacial contamination from the fabrication of devices)^{28,29}. To fulfil the full potential and high-end applications of next-generation integrated nonlinear electronic, optical and optoelectronic devices based on moiré superlattices (such as quantum nonlinear optics, compact parametric amplifiers, soliton-based optical communications and high-frequency rectifiers), it is of utmost importance and a prerequisite to achieve precise control of twist angle at the wafer scale, as well as minimizing impurities and strain disorder. Notable progress is being made in this direction, and the momentum is expected to persist, driving further advancements in the field. Efforts to develop new transfer techniques (for example, high-vacuum assembly, polymer-free dry transfer method and direct wafer-bonding) and automatic processes¹¹⁴, through integration with advanced machine learning algorithms, will greatly improve the reproducibility of large-scale device fabrication and accelerate the progress in emerging technological innovations. In addition, bottom-up epitaxial growth, by breaking the energy tendency and designing novel strategies (for example, kinetic control strategy and symmetry engineering strategy)^{115,116}, promises the prospect of batch production of homogeneous and twist-angle-controllable moiré superlattices with high reproducibility and scalability, laying a strong material foundation for the future integrated applications.

Looking ahead, the nonlinear physics of moiré superlattices has just begun, and endless opportunities are afoot. The future progression of the field will undoubtedly bring more surprises and further transform the landscape of basic scientific research and technological innovation in physics, materials science, quantum simulations, optical quantum technologies, communications and beyond.

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Author contributions

L.D. and Z.S. conceived the idea. L.D., Z.H., J.Z., F.Y., H.D., G.Z. and Z.S. led the writing. All authors reviewed and edited the paper before submission.

Competing interests

The authors declare no competing interests.

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